AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-08-07 | Entry window 7 August 2026 | 26 records in scope

26 RECORDS IN SCOPE (20 rejected, 4 metadata-only)

7 SENSING RECORDS (27% OF CORPUS)

8 SUBTOPIC CLUSTERS REPRESENTED

8 TIER-A RECORDS

16 EFFECT ESTIMATES EXTRACTED

Signal of the day

Two *EHP* cohorts published on the same day both find that *where the particle came from* carries more information than how much of it there is — and one of them shows the effect survives inside the household, where no monitoring network reaches.

Cantuaria et al. pooled **159,769** participants from the Danish Diet, Cancer and Health cohort and the Danish National Health Survey, reconstructed 10-year address histories, and decomposed exposure into *local traffic* and *other* sources for each of $PM_{2.5}$, NO_2, UFP and elemental carbon. Total $PM_{2.5}$ gave the largest association with incident COPD (**HR 1.11, 95% CI 1.05–1.17** per IQR), and the source split inverts between metrics: $PM_{2.5}$ *from sources other than local traffic* was the more harmful fraction, whereas for UFP and EC the local-traffic contribution dominated. A network built to resolve traffic gradients is therefore optimised for the wrong fraction of the $PM_{2.5}$ signal and the right one for UFP. Sassano et al. then supply the indoor mirror image in the Golestan Cohort (**50,045** adults, 724,064 person-years, 262 kidney deaths): biomass household fuel carried **HR 1.20 (1.04–1.37)** per 10 years of use for kidney-disease mortality, kerosene and gas were null — and the biomass estimate splits cleanly on ventilation, **1.19 (1.06–1.34)** without a chimney against **1.06 (0.95–1.18)** with one ($P_{\text{diff}} = 0.025$). The exposure that matters is not measured by any ambient monitor and is modified by a variable no exposure model carries. Both results point the same way: source and microenvironment, not concentration, are the unmeasured axes.

What changed today

- **Sixteen effect estimates with intervals — the largest set this watch has carried**, against six on 6 August and one on 5 August. Fourteen are PM or household-smoke estimates and two are co-pollutant. The CI drought that ran 2–5 August is now comprehensively over, and the reason is compositional: three large cohort or time-series papers (Cantuaria, Orr, Sassano) landed in the same window, each reporting a family of stratified estimates rather than a single headline. Three GBD burden records landed with them — India stillbirths, LRI and TB, ischaemic stroke at 20–54 — and contribute none, attributing fractions to air pollution as a risk factor rather than estimating an exposure–response.
- **A well-powered internal contradiction on wildfire smoke and influenza.** Orr et al. (*EHP*) find wildfire-season $PM_{2.5}$ associated with higher influenza risk in the four states reporting *laboratory-confirmed influenza* — Colorado **RR 1.067 (1.056–1.078)**, Arizona 1.061 (1.026–1.100), Oregon 1.049 (1.041–1.057), Montana 1.038 (1.013–1.063) per 10 $\mu g/m^3$ — and *no* effect in Nevada (1.005, 0.920–1.097) and a **protective** estimate in Washington (**0.884, 0.842–0.919**), the two states contributing only syndromic ILI counts. The split falls exactly along the outcome-ascertainment boundary, which is the most parsimonious explanation and one the paper raises but does not resolve. Read as an outcome-misclassification result rather than a geography result.
- **$PM_{2.5}$ composition beats $PM_{2.5}$ mass again, this time with a genotype interaction.** Li et al. (*Environ Int*) report that an IQR increase in all 15 measured $PM_{2.5}$ components jointly was associated with **10.0% higher MRI hepatic fat fraction** in 113 Los Angeles Latino adolescents with obesity, with elemental carbon, bromine, copper, potassium and lead as the drivers — a combustion/traffic signature — and effects concentrated in **PNPLA3 GG carriers**, who also showed 550 mediating metabolite features against 196 in non-carriers. Total mediation was significant *only* in GG carriers. Replication in 81 mixed-ethnicity young adults held for EC and Br. Small discovery cohort, single metropolitan area, and component estimates from a regional model rather than personal measurement.
- **Instrumentation: the day's best sensing content is largely un-summarisable.** Four Crossref by-ISSN deposits directly on the sensing axis — a low-cost PM sensor performance evaluation under Northeast China pollution episodes, an explainable-ML attribution of winter $PM_{2.5}$ drivers in China, surfactant effects on supermicron NaCl hygroscopic growth, and metal-ion/pH control of brown-carbon absorption — **carry no abstract in Crossref, Europe PMC or Semantic Scholar**, so they are listed on metadata and not summarised. What *is* summarisable is Jin et al. (*ACP*): three-or-more-day-old smoke transported into Missoula raised CO, $PM_{2.5}$ and total VOCs by factors of **2–8**, hourly $PM_{2.5}$ peaking at **120 $\mu g/m^3$**, tested against GEOS-Chem.
- **Two low-cost-sensor network records enter as backfill, both about deployment rather than accuracy.** DeMarsh et al. show a community PurpleAir network in Fresno County flagging four weeks of NAAQS-exceeding $PM_{2.5}$ at locations the regulatory network missed, while agreeing on concentration where the two overlap. Blanco-Villafuerte et al. describe a **176-monitor** national PurpleAir network in Peru feeding a hybrid LCS/satellite/CTM model that reached **R^2 0.88** in Lima. Both are Crossref-dated January–February 2026: backfill, off any trend line.

Corpus analytics

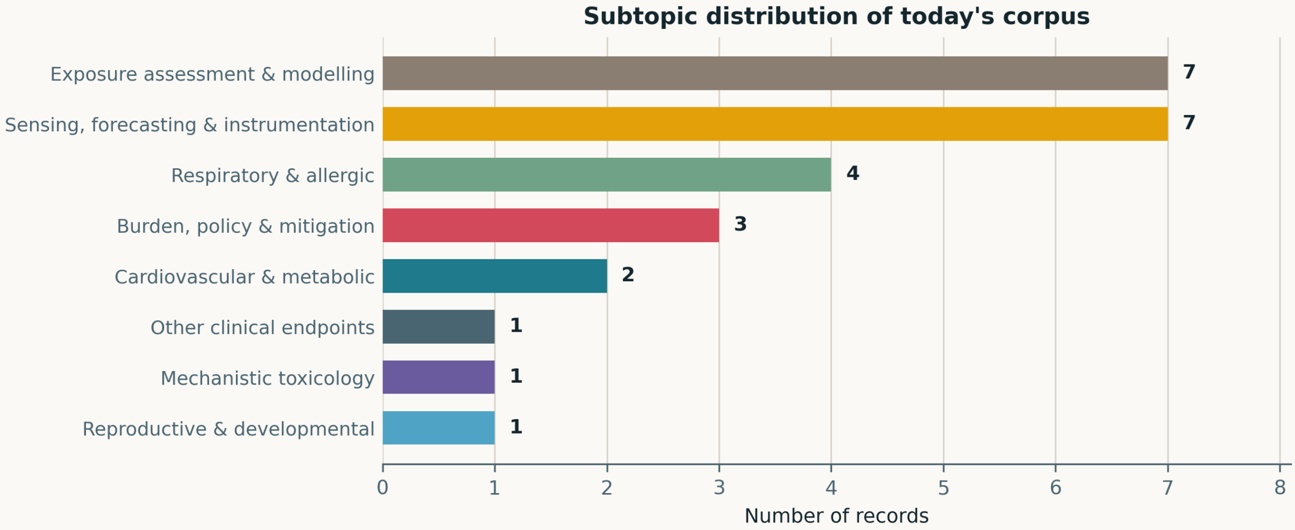
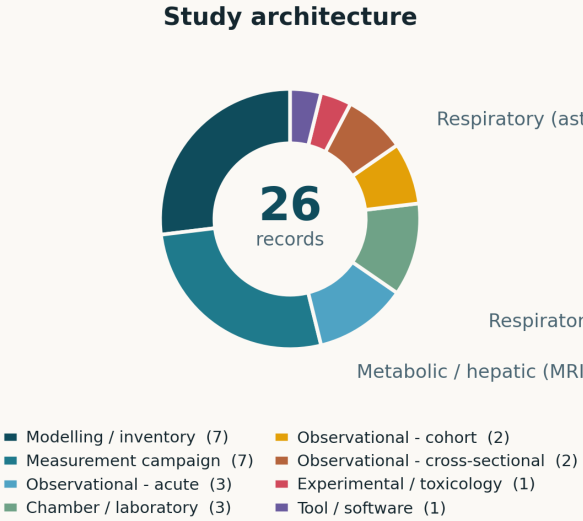


Fig. Subtopic assignment of the 26 in-scope records, single-label by dominant contribution. *Exposure assessment & modelling* and *Sensing, forecasting & instrumentation* tie at 7 each (27% apiece), so **54% of the window is methodological rather than epidemiological**. Respiratory & allergic follows at 4 and burden/policy at 3. Eight of the ten standing clusters are populated; *Neuro / mental health* and *Occupational & indoor* are empty as dominant labels, though the Sassano household-fuel cohort and the Hou filtration paper both sit adjacent to the latter.



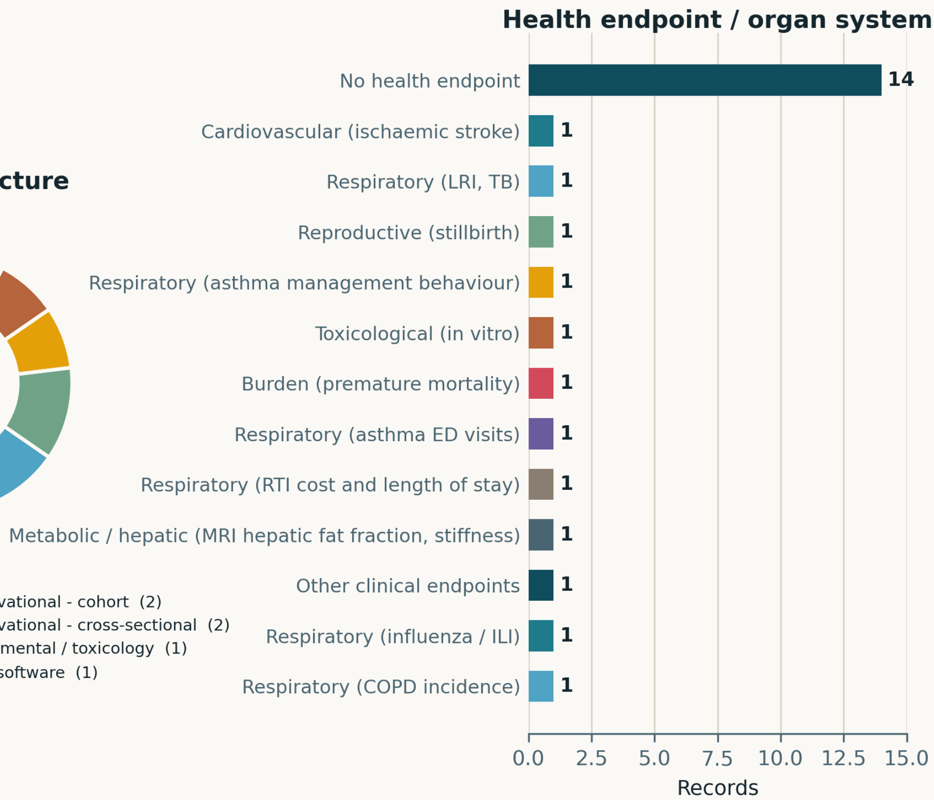


Fig. Left — study architecture. Modelling/inventory and measurement campaigns tie at 7 each, ahead of chamber/laboratory (3) and acute observational designs (3); only 2 records are cohort studies, but those 2 (Cantuaria, Sassano) carry 210,000 participants between them and supply half the day's effect estimates. Right — health endpoint by organ system. **Fourteen of the 26 records carry no human health endpoint**, and the twelve that do are spread one per endpoint with no two records sharing a label — consistent with a methods-heavy window and with the register's breadth. There is no trial or intervention record in this window.

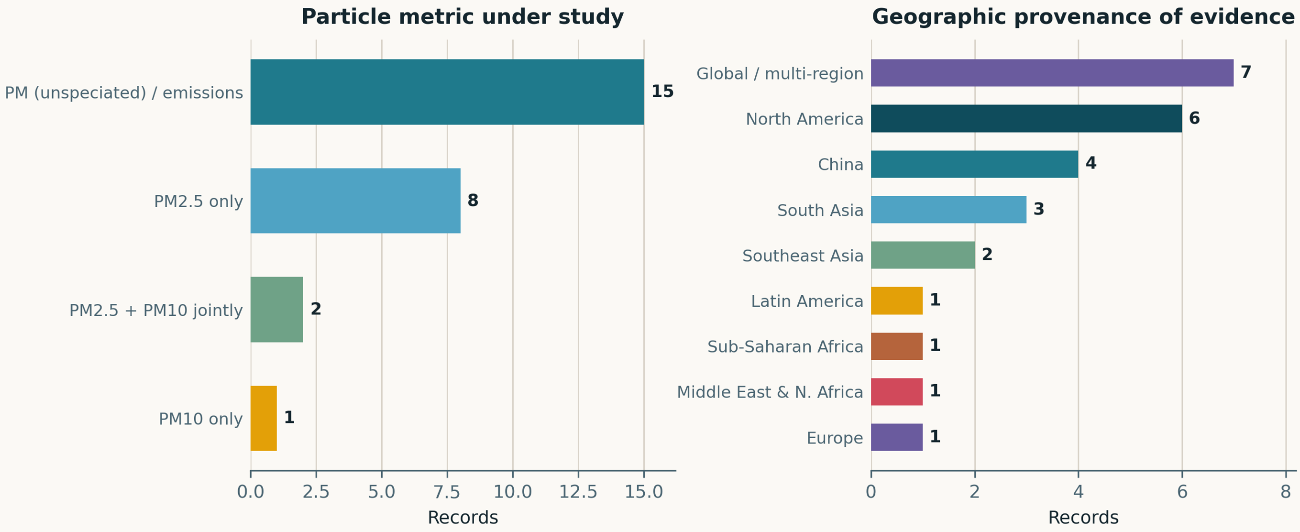


Fig. Left — particle metric. Eight records use $PM_{2.5}$ alone, two pair it with PM_{10} and one uses PM_{10} alone. The 15-record unspeciated/emissions bar is genuine heterogeneity, not missing detail: it holds $PM_{0.3}$ filtration efficiency, speciated organosulfates, brown-carbon absorption, hygroscopic growth factors, trace-metal bioaccumulation, household fuel type and three GBD risk-factor attributions. Right — geographic provenance. North America leads (6), then the global/multi-region group (7, which includes the four laboratory and computational records placed there by intent) and China (4). **South Asia contributes 3 and Sub-Saharan Africa 1.**

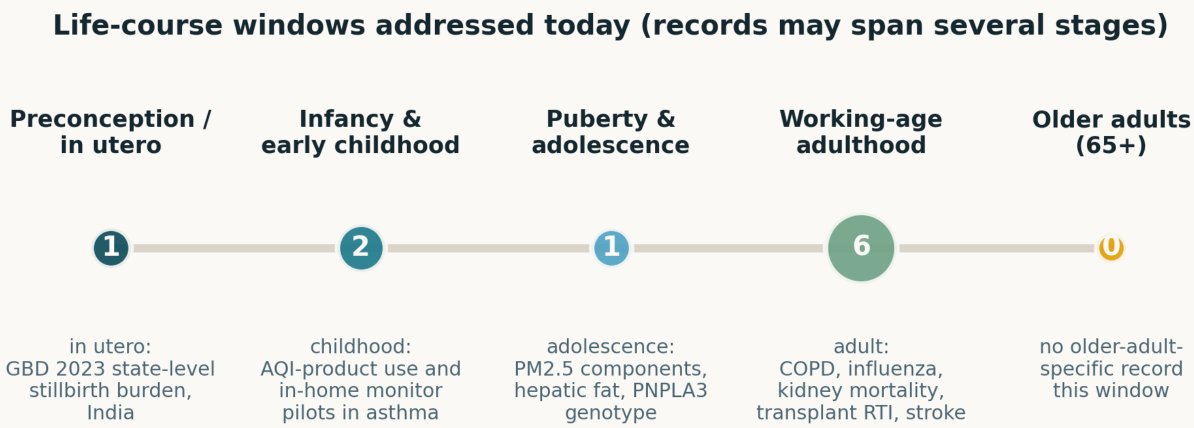


Fig. Life-course windows addressed; counts are non-exclusive. Ten of 26 records carry a human life-course window and adults dominate (6). **The adolescent window is populated for the first time in five issues** by Li et al.'s PNPLA3 hepatic-fat cohort. The **older-adult window is empty for a fourth consecutive issue**, which remains the most conspicuous standing gap given that group's cardiovascular attributable burden.

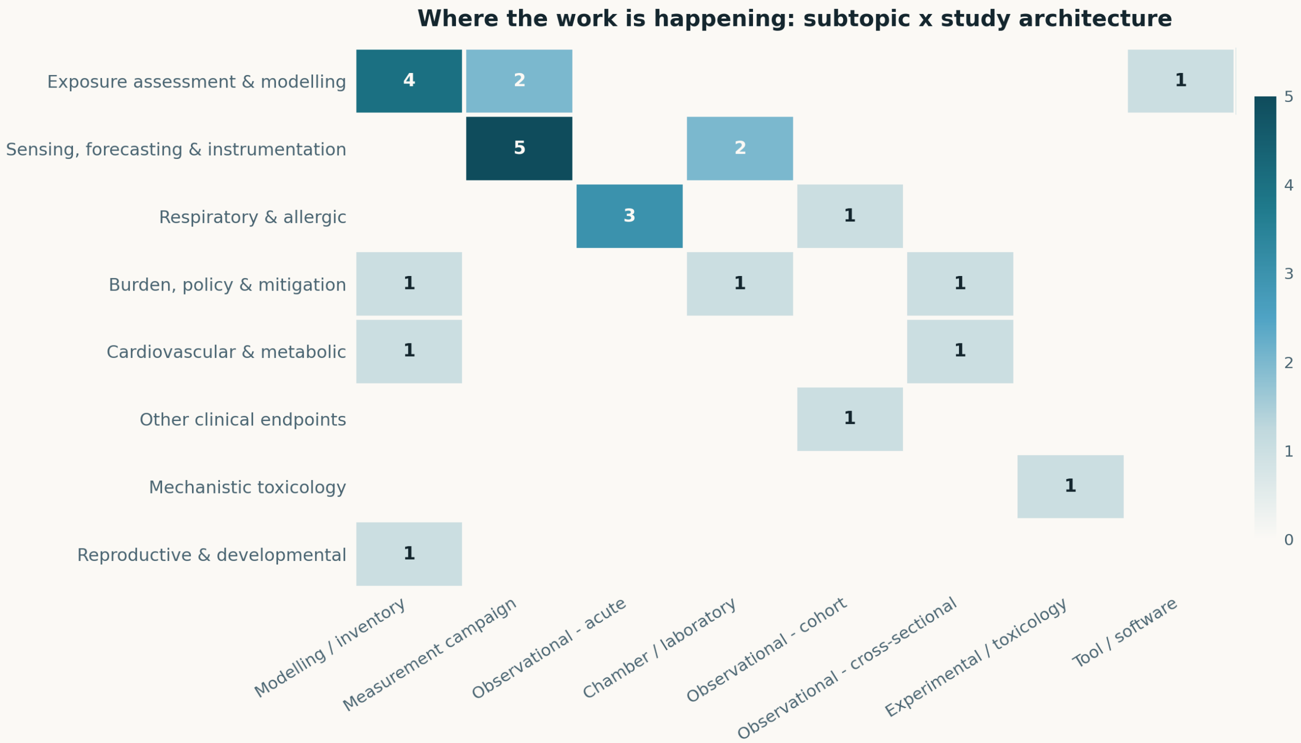


Fig. Subtopic × study-architecture cross-tabulation. The *Sensing* row is again confined to measurement campaigns and laboratory work with **no cell in any observational-epidemiology column**, unbroken in every issue since 27 July. The converse also holds: the two cohort studies sit entirely in the respiratory and other-clinical rows. The one crossing cell is Orr et al.'s in-home monitor pilot, which is a sensing record with a clinical population attached.

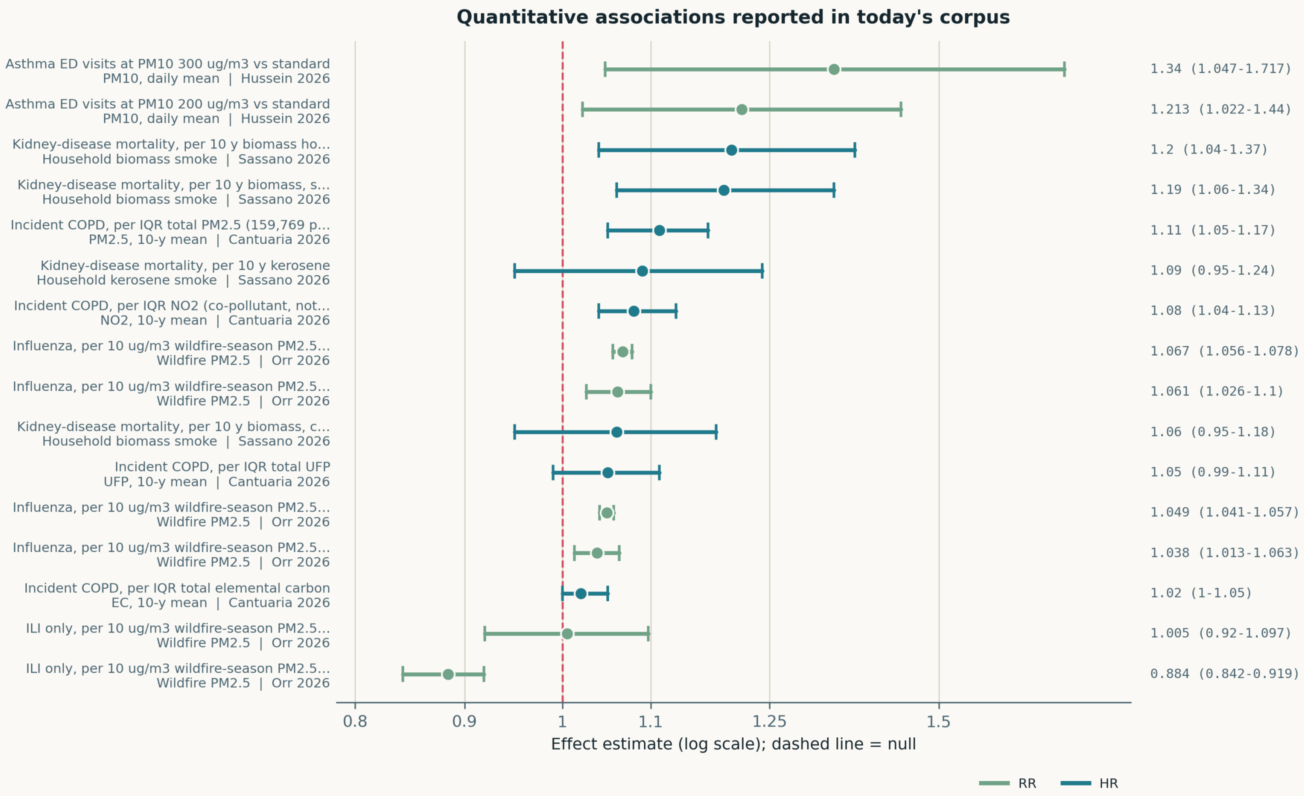


Fig. All sixteen effect estimates carrying an interval, on a common logarithmic axis. **Exposure contrasts are heterogeneous and the estimates are not mutually comparable**: Cantuaria is per IQR of a 10-year mean, Orr is per 10 $\mu g/m^3$ of wildfire-season $PM_{2.5}$, Sassano is per 10 years of household fuel use, and Hussein is a threshold contrast against the Malaysian standard. **The Washington ILI estimate (0.884, 0.842–0.919) is the only point below the null** and is best read as outcome-misclassification rather than protection — it and Nevada are the two syndromic-ILI states. One entry is a co-pollutant estimate (NO_2, Cantuaria) and is marked as such; the remaining fifteen are PM or household-smoke exposures.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Respiratory & allergic 4 records

Cantuaria et al. 2026 Environ Health Perspect

Source-specific $PM_{2.5}$, UFP, EC and NO_2 and incident COPD in two pooled Danish cohorts

159,769 participants from the Diet, Cancer and Health cohort and the Danish National Health Survey, with complete residential address histories driving 10-year time-weighted exposures split into local-traffic and other-source contributions; Cox models adjusted for demographic, socioeconomic and lifestyle factors including smoking. Per IQR of total exposure: **$PM_{2.5}$ HR 1.11 (1.05–1.17)**, NO_2 1.08 (1.04–1.13), UFP 1.05 (0.99–1.11), EC 1.02 (1.00–1.05). **The source split inverts across metrics** — non-traffic $PM_{2.5}$ was the more harmful fraction while traffic-derived UFP and EC dominated theirs. Associations were stronger in women, never-smokers and prior asthma. Residual confounding by smoking intensity is the live threat given COPD's dominant cause. Directly relevant to siting: a traffic-gradient network resolves the wrong $PM_{2.5}$ fraction and the right UFP one.

doi: 10.1021/EHP.6c00128

Orr et al. 2026 (EHP) Environ Health Perspect

Wildfire-season $PM_{2.5}$ and influenza / ILI risk across six western US states

Generalised linear distributed-lag models on state health-department influenza and ILI counts from Arizona, Colorado, Montana, Nevada, Oregon and Washington, 2010–2019, testing both preceding-wildfire-season exposure on the following flu season and short-term exposure within it. Per 10 $\mu g/m^3$: **CO 1.067 (1.056–1.078), AZ 1.061 (1.026–1.100), OR 1.049 (1.041–1.057), MT 1.038 (1.013–1.063)** — all four states with laboratory-confirmed influenza data — against **NV 1.005 (0.920–1.097) and WA 0.884 (0.842–0.919)**, the two with syndromic ILI only. **The effect boundary coincides exactly with the outcome-ascertainment boundary**, so the geographic reading is almost certainly the wrong one. Ecological design, no individual exposure, and wildfire-attributable $PM_{2.5}$ is itself a modelled quantity.

doi: 10.1021/EHP.6c00326

Hussein et al. 2026 J Glob Health

PM_{10} and emergency asthma presentations in Petaling Jaya, Malaysia

Four years of ED records (Jan 2014 – Apr 2017) from a Klang Valley hospital against pollutant and meteorological data from nearby stations; generalised additive models plus distributed lag non-linear models for delayed and non-linear PM_{10} effects. Asthma exacerbations were 2.3% of all ED visits. Relative to the Malaysian standard, daily mean PM_{10} of **200 $\mu g/m^3$ was associated with a 21.3% increase (95% CI 2.2–44.0)** and **300 $\mu g/m^3$ with 34.0% (4.7–71.7)**; the cumulative lag 0–7 exposure–response turns upward from about 150 $\mu g/m^3$, and the lag structure shifts from days 2–3 at 100 $\mu g/m^3$ to same-day at $\geq$200. Single hospital, single monitoring catchment, wide intervals — the authors correctly label the associations exploratory. The threshold shape is the transferable part: it argues for alert triggers well below current national standards.

doi: 10.7189/jogh.16.04249

Jiang et al. 2026 J Cent South Univ Med Sci

Short-term PM and respiratory-infection hospitalisation burden in kidney transplant recipients

Retrospective time-series over 874 kidney transplant recipients hospitalised for respiratory tract infection at the Third Xiangya Hospital, 2017–2023, with high-resolution daily exposure to six pollutants over the 7 days before symptom onset and DLNMs for lagged and cumulative associations. **$PM_{2.5}$ showed the strongest association with burden: at 160 $\mu g/m^3$ cumulative exposure, hospitalisation cost rose by 9,544.5 yuan (95% CI 2,548.8–21,637.7)**; PM_{10} showed a dose-dependent trend on length of stay (RR 1.63 at 300 $\mu g/m^3$) whose interval, 0.18–15.94, is so wide it carries no information and is deliberately excluded from the forest plot. SO_2 above 75 $\mu g/m^3$ also raised burden; CO, NO_2 and O_3 were null. Single centre, immunosuppressed population, ambient exposure assigned without residential history. Economic rather than clinical endpoints, which is unusual and useful.

doi: 10.11817/j.issn.1672-7347.2026.260089

Sensing, forecasting & instrumentation 7 records

Jin et al. 2026 Atmos Chem Phys

Emissions, chemistry and health impact of aged wildfire smoke over Missoula, Montana

Hourly surface observations of $PM_{2.5}$, CO, NO_x, O_3 and **75 speciated VOCs** through a strong 2020 smoke episode, used as a direct test of how wildfire emissions, ageing chemistry and health effects are represented in GEOS-Chem. Smoke three or more days old, transported from California and the Pacific Northwest, raised CO, $PM_{2.5}$ and total measured VOCs by **factors of 2–8**, with hourly maxima of 800 ppb CO and **120 $\mu g/m^3$ $PM_{2.5}$**. The value here is the speciated VOC suite alongside mass: it is the composition record against which a mass-only low-cost network's health interpretation has to be judged, and it is one city and one episode, so the transport chemistry is not generalisable without replication at other receptor sites.

doi: 10.5194/acp-26-11047-2026

Orr et al. 2026 (JMIR) JMIR Pediatr Parent

In-home digital air quality monitors in families of children with asthma: feasibility and acceptability

Pilot mixed-methods study, **N = 20** families, combining interviews and surveys to assess whether a consumer in-home air quality monitor can be implemented in paediatric asthma care. Use was reported as both feasible and acceptable, with **affordability identified as the principal barrier to clinical adoption**. The report is very short and carries no exposure data, no adherence metric and no asthma outcome, so it establishes only that the deployment step is not the bottleneck. That is still the piece of the low-cost-sensing translation chain that is least often evidenced — most LCS literature stops at accuracy and never asks whether a clinical population will host the device. Needs a powered trial with objective device-uptime data before it can bear weight.

doi: 10.2196/92070

Adeniji et al. 2026 PLOS Glob Public Health

Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria

Hourly $PM_{2.5}$, PM_{10}, NO_2, SO_2, CO and O_3 over 2012–2023 with six meteorological covariates. **$PM_{2.5}$ and PM_{10} dominate and persistently exceed WHO guidelines**, peaking in the December–March Harmattan; regression against temperature gives R^2 0.48–0.52 with negative coefficients on rainfall and humidity, consistent with thermal stagnation and wet scavenging. The aggregate AQI sits at 39–51 (“moderate”) while the health-risk-adjusted index exceeds 90 seasonally — **a 2-fold divergence between the two indices at identical concentrations**, which is the paper’s real contribution and an argument for HAQI-style reporting on any network serving a PM-dominated airshed. Single station for a metropolitan area, and the provenance and calibration history of a 12-year record in a data-sparse setting are not documented.

doi: 10.1371/journal.pgph.0006683

DeMarsh et al. 2026 (backfill) Atmosphere

Community PurpleAir network versus regulatory monitors for $PM_{2.5}$ in Fresno County

The SJV-CAIR network at UC Merced, built with community-based organisations, compared against regulatory monitoring across Fresno County for June–July 2023. Community monitors returned **concentration estimates comparable to regulatory instruments** where the two co-locate, but identified **four weeks in which some locations episodically exceeded the NAAQS that the regulatory network did not flag**, and interpolation maps built with the denser network resolved spatial structure the regulatory-only map did not. Two summer months in one county is a short and seasonally atypical window for the San Joaquin Valley, and no formal co-location correction is reported. Carried as backfill — Crossref `created` 11 February 2026, outside this entry window — and must be excluded from weekly trend lines.

doi: 10.3390/atmos17020187

Yukhymchuk et al. 2026 Aerosol Air Qual Res

Low-cost particulate matter sensor performance under polluted conditions in Northeast China

Crossref deposit of 7 August with **no abstract in Crossref, Europe PMC or Semantic Scholar**. Carried on title, journal and DOI only; **no summary is offered because none can be written without inventing content**. On title alone this is the single most directly relevant record of the window — LCS performance characterisation at the high concentrations where light-scattering response is least well constrained — and it should be revisited as soon as an abstract appears.

doi: 10.1007/s44408-026-00159-6

Sanders et al. 2026 Aerosol Sci Technol

Nonionic surfactant composition and hygroscopic growth of supermicron NaCl particles

Crossref deposit of 7 August, **metadata only — no abstract deposited anywhere in the retry chain**, so no summary is offered. Recorded here because surfactant control of hygroscopic growth is the physical mechanism underlying humidity correction in light-scattering PM sensors, and the supermicron size range is where nephelometric response is most sensitive to growth factor.

doi: 10.1080/02786826.2026.2703042

Huang et al. 2026 Atmos Pollut Res

Metal ions and pH as controls on the optical properties of atmospheric brown carbon

Crossref deposit of 7 August, **metadata only — no abstract retrievable**, so no summary is offered. Logged because brown-carbon absorption is the aerosol optical property that most directly confounds the fixed refractive-index assumptions built into low-cost optical particle counters.

doi: 10.1016/j.apr.2026.103167

Exposure assessment & modelling 7 records

Myat et al. 2026 Environ Pollut

Ground, satellite and model assessment of $PM_{2.5}$ and attributable mortality in Myanmar

Multi-source assessment for a least-developed country with minimal regulatory monitoring: two Yangon ground stations 2019–2024, TROPOMI HCHO and NO_2 nationally, and a generalised additive model for meteorological dependence. Mean $PM_{2.5}$ was **24.4 μg/m^3 (SD 18.6)**, rising to **39.4 μg/m^3** in the November–February winter, with sulfate the leading measured component year-round and nitrate and Ca^{2+} seasonally. Compliance with the 5 μg/m^3 WHO guideline would have averted **2,106 (95% CI 345–3,818) premature deaths annually** among adults $\geq$30 in Yangon over 2020–2021. TROPOMI HCHO/NO_2 ratios indicate NO_x- sensitive ozone chemistry nationally. **Two stations cannot represent a city of five million**, and the health impact assessment inherits every assumption of the imported concentration–response function.

doi: 10.1016/j.envpol.2026.128918

Cai et al. 2026 (ACP) Atmos Chem Phys

Molecular characteristics and formation pathways of particulate organosulfur compounds in Shanghai

Paired urban and suburban Shanghai sampling with high-resolution mass spectrometry, resolving **1,964, 1,914 and 2,689 organosulfur molecular formulas** across the sampled conditions and separating spatial, seasonal and day–night variation in formation pathway. Organosulfates are ubiquitous but their production regime is not: the comparison across contrasting sites is what isolates the pathways rather than the inventory itself. The limitation that matters for downstream use is that molecular formula assignment from exact mass does not fix structure, so pathway attribution is inferential. Relevant to sensing as a concrete demonstration that a large, chemically distinct fraction of urban $PM_{2.5}$ mass is invisible to any mass-only instrument.

doi: 10.5194/acp-26-11067-2026

Blanco-Villafuerte et al. 2026 (backfill) Curr Environ Health Rep

Hybrid low-cost-sensor, satellite and chemical-transport modelling of ambient $PM_{2.5}$ in Peru

Review of national $PM_{2.5}$ exposure-model construction in LMICs, anchored on the NIH-funded Peru GeoHealth Hub’s **176-monitor PurpleAir network** — 62.5% urban, 37.5% rural, across all 24 regions, established with national environmental agencies. The hybrid model fuses LCS, satellite retrievals, chemical transport output and machine learning; initial application to Lima reached **R^2 = 0.88** against regulatory monitors for 2010–2023, with extension to daily national 5 km^2 fields for 2024–2026 in progress. R^2 against regulatory monitors in a single city is a weak validation for a national product, and the review is written by the network’s own investigators. The deployment-ratio and institutional-integration detail is the transferable content. Crossref `created` 5 January 2026 — backfill, exclude from trend lines.

doi: 10.1007/s40572-025-00508-4

Zheng et al. 2026 Atmos Environ

Spatially heterogeneous drivers of winter $PM_{2.5}$ in China from synchronous observation and explainable ML

Crossref deposit of 7 August, **metadata only — Elsevier deposits no abstract to Crossref and the record is not yet in Europe PMC, PubMed or Semantic Scholar**. No summary is offered. Logged because synchronous multi-site observation paired with explainable machine learning is the design that the sensing literature most often lacks: it is the combination that could distinguish genuine spatial heterogeneity in drivers from instrument heterogeneity across a network.

doi: 10.1016/j.atmosenv.2026.122285

Rai et al. 2026 Environ Monit Assess

Land surface temperature, land cover and aerosol optical depth over Ahmedabad, 1993–2023

Multi-sensor remote sensing: Landsat LST and urban thermal field variance index for 1993, 2002, 2014 and 2023, with MODIS AOD and LST from 2002–2023 for the diurnal response. **Urban extent grew by 171.15 km^2 over 30 years** with sharp losses of fallow and vegetated land, and the thermal environment polarised — fallow land carries the worst thermal comfort, water and vegetation the best, and NDVI/NDBI track the expected cooling and warming signs. **The AOD–LST correlation is dual-signed**, which the paper reports but does not mechanistically decompose into aerosol direct radiative effect versus co-varying surface change; that confounding is the main threat to the inference. Read as a heat-planning input rather than a PM exposure product — AOD here is a columnar quantity never converted to surface concentration.

doi: 10.1007/s10661-026-15748-6

Meena et al. 2026 Environ Pollut

Transplanted lichen trace-metal accumulation along a Jaipur urban gradient

Phaeophyscia hispidula thalli collected from a wildlife-sanctuary reference site and transplanted to ten urban and one reference location in Jaipur for 90 days over the March–June 2024 pre-monsoon, with Zn, Pb, Cd, Cr, Cu and Ni by ICP-MS and chlorophyll and carotenoid concentrations by spectrophotometry. Urban sites accumulated more metal, and **accumulation tracked chlorophyll loss while carotenoids rose** at several exposed sites, consistent with a photoprotective adjustment. PCA and hierarchical clustering resolved a spatial gradient. **No co-located instrumental PM measurement means the biomonitor is never calibrated against a concentration**, so this yields a relative ranking of sites, not an exposure estimate. Useful as an ultra-low-cost spatial screening layer where no network exists; not a substitute for one.

doi: 10.1016/j.envpol.2026.128919

Temu et al. 2026 Environ Sci Pollut Res

A six-domain framework for lichen biomonitoring including symbiont and microbiome response

Narrative synthesis proposing that lichen biomonitoring be extended from the classical triad of pollutant accumulation, physiological injury and community composition to six domains, adding functional traits, symbiotic interaction and lichen-associated microbiome dynamics as **earlier-responding indicators** than bulk accumulation. The argument that microbiome shifts precede visible physiological injury is plausible and is the part worth testing, but the review supplies **no quantitative comparison of detection limits or response latency across the six domains**, so the central claim of earlier warning is asserted rather than demonstrated. The authors name geographic bias, species-specific response and absent methodological standardisation as open problems — the same standardisation gap that limits low-cost sensor networks, and for structurally similar reasons.

doi: 10.1007/s11356-026-38123-x

Cardiovascular & metabolic 2 records

Li et al. 2026 (Environ Int) Environ Int

$PM_{2.5}$ constituents, hepatic fat and stiffness in youth, modified by PNPLA3 genotype

113 Los Angeles Latino adolescents with obesity, MRI-measured hepatic fat fraction and liver stiffness, serum metabolomics and lipidomics, and visit-year mean exposure to **15 $PM_{2.5}$ components** at residence; linear regression plus g-computation for mixture effects with PNPLA3 genotype as modifier, and moderated mediation. A simultaneous IQR increase across all components was associated with **10.0% higher hepatic fat fraction**, with elemental carbon, bromine, copper, potassium and lead as leading contributors — a combustion and traffic signature. **Effects concentrated in PNPLA3 GG carriers**, who showed 550 mediating metabolite features against 196, broader pathway enrichment, and the only significant total mediation. Replication in 81 mixed-ethnicity young adults held for EC and Br. Small discovery sample, cross-sectional, modelled rather than personal exposure — the genotype interaction is the hypothesis, not the estimate.

doi: 10.1016/j.envint.2026.110438

Zhao et al. 2026 Medicine (Baltimore)

Ischaemic stroke burden and attributable risk factors in adults aged 20–54, 1990–2021

GBD 2021 extraction for ischaemic stroke in the 20–54 age band, with age-period-cohort models and population-attributable fractions by risk factor. In 2021 there were **16,301,644 prevalent and 1,324,743 incident cases** (432.46 and 35.14 per 100,000), 105,859 deaths and 6.97 million DALYs. **Prevalence and incidence rose over the period while mortality and DALYs fell**, with burden concentrated in high-middle SDI regions, Eastern Europe, Central Asia and parts of North Africa and the Middle East. Air pollution enters only as one of the GBD attributable risk factors, so this contributes no exposure–response estimate and inherits every GBD modelling assumption, including sparse primary data in the highest-burden regions. Carried for the age-band signal: young- adult stroke burden is rising in exactly the regions where PM exposure is highest and monitoring is thinnest.

doi: 10.1097/MD.0000000000050104

Burden, policy & mitigation 3 records

Hou et al. 2026 J Hazard Mater

Bimodal polysulfonamide fibrous aerogel for high-temperature $PM_{0.3}$ filtration

Co-electrospun aerogel combining thermally stable PSA/PAN/SiO_2 coarse fibres with pure PSA fine fibres in a 3D curly network at **>98% porosity**, formed via moisture-induced phase separation and jet whipping from the mechanical mismatch between components. **$PM_{0.3}$ filtration efficiency 98.5% at ∼84 Pa pressure drop**, retained at **92.31% and 73 Pa after 12 h at 250 °C**, with acid and alkali resistance and useful thermal insulation. The gap in the report is loading behaviour: no long-duration dust-holding capacity or efficiency-versus-loading curve is given, and that is what determines service life in a real industrial stack. Relevant to source-side mitigation rather than monitoring, and to the ultrafine fraction that mass-based networks under-weight.

doi: 10.1016/j.jhazmat.2026.143218

Han et al. 2026 AJPM Focus

Air quality index product use versus clinician discussion among US children with asthma

Questionnaire administered inside the Transcript/Omics for Precision Medicine in Asthma study, youth aged 6–21 recruited 2023–2025, with bivariate testing. **75% of caregivers were aware of an AQI or alert, 64% checked it before going outdoors and 73% reported a behaviour change** — but the delivery channel is consumer, not clinical or public: a phone weather app in 75% of cases, broadcast or social alerts 44%, websites 24%, county services only **14%**. Only 41% had ever discussed air pollution with a clinician and 35% received poor-air-quality recommendations. Self-report with no objective behaviour measure and a single research cohort limit generalisability. The finding that matters for network design: **the operative last mile is a commercial weather app, and public monitoring agencies reach one family in seven**.

doi: 10.1016/j.focus.2026.100504

Caikai et al. 2026 J Infect Dev Ctries

Risk-factor-attributable burden of respiratory infection and tuberculosis, 1990–2021

GBD 2021 across 204 countries, decomposing deaths, DALYs and age-standardised rates for respiratory infection and TB across nine risk factors by age, sex and SDI, with APC and AAPC trend estimation. Age-standardised mortality attributable to the traditional risks fell steadily — **air pollution −3.18% annually**, malnutrition −4.46%, alcohol −3.35% — while metabolic-risk-attributable deaths rose in absolute terms despite falling rates. Tobacco-attributable mortality was **3.6-fold higher in men**, and children under 5 remain the group most exposed to environmental risks. Air pollution is a GBD attributable fraction here, not a measured exposure–response, so this is a trend context record: the attributable burden is falling on rate, which the accompanying absolute counts do not support as a success narrative.

doi: 10.3855/jidc.21667

Other clinical endpoints 1 record

Sassano et al. 2026 Environ Health Perspect

Household solid-fuel use and kidney-disease mortality in the Golestan Cohort

50,045 adults aged 40–75 enrolled 2004–2008 in northeastern Iran and followed to April 2023, with validated fuel-use questionnaires and Cox models; 262 kidney deaths over **724,064 person-years**. Per 10 additional years of household biomass use for cooking or heating, **HR 1.20 (95% CI 1.04–1.37)** for kidney-disease mortality; kerosene 1.09 (0.95–1.24) and gas 1.00 (0.86–1.16) were null. **The biomass effect splits on ventilation** — 1.19 (1.06–1.34) without a chimney against 1.06 (0.95–1.18) with one, $P_{\text{diff}} = 0.025$ — which is the strongest evidence in the paper that the exposure is the smoke rather than a correlate of poverty, though confounding by socioeconomic position is not fully excluded. **No PM measurement of any kind**: fuel type and duration are the entire exposure model, and the ventilation interaction is precisely the variable no ambient network can observe.

doi: 10.1021/EHP.6c00520

Mechanistic toxicology 1 record

Wu et al. 2026 (JHM) J Hazard Mater

TNF-α and EGF mediate endothelial activation downstream of $PM_{2.5}$ alveolar injury

3D alveolar–vascular barrier constructed from BEAS-2B epithelium, THP-1 macrophages and EA.hy926 endothelium, exposed to **100 μg/mL SRM2786 $PM_{2.5}$ for 24 h**. $PM_{2.5}$ entered epithelial cells by endocytosis, produced mitochondrial damage and apoptosis, and raised TNF-α, EGF, IL-8, TGF-α and IL-1β; extracellular **TNF-α and EGF crossed the barrier and activated endothelial PI3K–Akt**, with Nrf2 and CYP1A1 induction marking progressive oxidative stress. The model supplies a mechanistic route from local alveolar injury to systemic endothelial dysfunction. The single exposure concentration is far above any ambient equivalent and there is no dose–response, so the result establishes the pathway's existence, not its relevance at realistic exposures — the standard and unresolved weakness of this literature.

doi: 10.1016/j.jhazmat.2026.143204

Reproductive & developmental 1 record

GBD 2023 India Stillbirth Collaborators 2026 Lancet Reg Health Southeast Asia

State-level stillbirth rates in India at ≥22 and ≥28 weeks, GBD 2023

First GBD estimation of stillbirth rate under the WHO ICD-11 ≥22-week definition alongside the conventional ≥28-week cutoff, disaggregated to 31 Indian geographic units. **122 data sources and 2,684 location-year observations** were modelled by spatiotemporal Gaussian process regression on the stillbirth-to-neonatal-mortality ratio, anchored on GBD 2023 fertility and all-cause neonatal mortality, with 95% uncertainty intervals and a secondary analysis against SDG3 indicators. **India carries the largest global stillbirth burden at both cutoffs**, and the ≥28-week convention materially understates it. Air pollution enters only through GBD's risk-factor architecture, so no PM exposure–response is estimated here. Carried because state-level disaggregation is the resolution at which a PM–stillbirth analysis could be attempted, and because it defines the denominator such an analysis would use.

doi: 10.1016/j.lansea.2026.100819

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Li et al. (Environ Int)	$PM_{2.5}$ constituents, hepatic fat and stiffness in youth, and effect modification by PNPLA3 genotype	*Environ Int*	Cross-sectional imaging	15 $PM_{2.5}$ components, visit-year mean at residence	North America
Zhao et al.	Ischaemic stroke burden and attributable risk factors in adults aged 20-54, 1990-2021	*Medicine (Baltimore)*	Burden estimation (GBD)	Air pollution as GBD risk factor (PAF)	Global / multi-region
Reproductive & developmental					
GBD 2023 India Stillbirth Collab.	State-level stillbirth rates in India at >=22 and >=28 weeks, GBD 2023	*Lancet Reg Health Southeast Asia*	Burden estimation (GBD)	Risk-factor attribution incl. ambient and household PM	South Asia
Respiratory & allergic					
Cantuaria et al.	Source-specific $PM_{2.5}$, NO2, UFP and elemental carbon and incident COPD in two pooled Danish cohorts	*Environ Health Perspect*	Prospective cohort	$PM_{2.5}$, UFP, EC, NO2 (10-y time-weighted, address history)	Denmark
Hussein et al.	PM_{10} and emergency visits for asthma exacerbation in Petaling Jaya, Malaysia	*J Glob Health*	Time-series / case-crossover	PM_{10}, daily mean, DLNM lag 0-7	Southeast Asia
Jiang et al.	Short-term $PM_{2.5}$, PM_{10} and SO2 and respiratory-infection hospitalisation burden in kidney transplant recipients	*J Cent South Univ Med Sci*	Time-series / case-crossover	$PM_{2.5}$, PM_{10}, SO2, NO2, CO, O3; 7-day pre-symptom cumulative	China
Orr et al. (EHP)	Wildfire-season $PM_{2.5}$ and influenza / influenza-like-illness risk across six western US states	*Environ Health Perspect*	Distributed-lag time-series	Wildfire-specific $PM_{2.5}$ (per 10 $\mu g/m^3$)	North America
Mechanistic toxicology					
Wu et al. (JHM)	TNF-alpha and EGF mediate endothelial activation downstream of $PM_{2.5}$ alveolar injury in a 3D barrier model	*J Hazard Mater*	Toxicological (in vitro)	SRM2786 $PM_{2.5}$, 100 ug/mL, 24 h	Computational / laboratory
Sensing, forecasting & instrumentation					
Adeniji et al.	Twelve years of hourly criteria-pollutant and meteorological coupling in Abeokuta, Nigeria	*PLOS Glob Public Health*	Monitoring-record analysis	$PM_{2.5}$, PM_{10}, NO2, SO2, CO, O3, hourly 2012-2023	Sub-Saharan Africa
DeMarsh et al. (backfill)	Community PurpleAir network versus regulatory monitors for $PM_{2.5}$ characterisation in Fresno County	*Atmosphere*	Sensor co-location	PurpleAir PA-II $PM_{2.5}$ vs regulatory FEM	North America
Huang et al.	Metal ions and pH as controls on the optical properties of atmospheric brown carbon	*Atmos Pollut Res*	Chamber / laboratory	Brown carbon absorption spectra	Computational / laboratory
Jin et al.	Emissions, chemistry and health impact of aged wildfire smoke in Missoula, Montana	*Atmos Chem Phys*	Measurement campaign	Hourly $PM_{2.5}$, CO, NOx, O3 and 75 speciated VOCs; GEOS-Chem	North America
Orr et al. (JMIR)	Feasibility and acceptability of a digital in-home air quality monitor in families of children with asthma	*JMIR Pediatr Parent*	Sensor deployment (pilot)	Indoor PM (consumer digital monitor)	North America
Sanders et al.	Nonionic surfactant composition and hygroscopic growth of supermicron NaCl particles	*Aerosol Sci Technol*	Chamber / laboratory	Hygroscopic growth factor, supermicron NaCl	Computational / laboratory
Yukhymchuk et al.	Low-cost particulate matter sensor performance under polluted conditions in Northeast China	*Aerosol Air Qual Res*	Sensor co-location	Low-cost PM sensor vs reference	China
Exposure assessment & modelling					
Blanco-Villafuerte et al. (backfill)	Hybrid low-cost-sensor, satellite and CTM modelling of ambient $PM_{2.5}$ in Peru	*Curr Environ Health Rep*	Machine-learning model	LCS + satellite AOD + CTM hybrid, 5 km daily	Latin America
Cai et al.	Molecular characteristics and formation pathways of particulate organosulfur compounds in Shanghai	*Atmos Chem Phys*	Measurement campaign	Particulate organosulfates, FT-ICR MS	China
Meena et al.	Transplanted lichen trace-metal accumulation and pigment response along a Jaipur urban gradient	*Environ Pollut*	Biomonitoring transplant	Zn, Pb, Cd, Cr, Cu, Ni by ICP-MS; chlorophyll / carotenoid	South Asia
Myat et al.	Ground, satellite and model assessment of $PM_{2.5}$ dynamics and attributable mortality in Myanmar	*Environ Pollut*	Physical exposure model	$PM_{2.5}$ speciated, TROPOMI HCHO/NO2, GAM	Southeast Asia

First author	Focus	Journal	Design	Metric	Region
Rai et al.	Land surface temperature, land cover and aerosol optical depth over Ahmedabad, 1993-2023	*Environ Monit Assess*	Satellite remote sensing	MODIS AOD (2002-2023), Landsat LST	South Asia
Temu et al.	A six-domain framework for lichen biomonitoring including symbiont and microbiome response	*Environ Sci Pollut Res*	Literature review	Not applicable (framework)	Global / multi-region
Zheng et al.	Spatially heterogeneous drivers of winter $PM_{2.5}$ in China from synchronous observations and explainable machine learning	*Atmos Environ*	Machine-learning model	Winter $PM_{2.5}$, synchronous multi-site observation	China
Burden, policy & mitigation					
Caikai et al.	Risk-factor-attributable burden of respiratory infection and tuberculosis, 1990-2021	*J Infect Dev Ctries*	Burden estimation (GBD)	Air pollution as GBD risk factor (ASMR, ASDR)	Global / multi-region
Han et al.	Use of air quality index products versus clinician discussion among US children with asthma	*AJPM Focus*	Cross-sectional survey	Self-reported AQI product use	North America
Hou et al.	Bimodal polysulfonamide fibrous aerogel for high-temperature PM0.3 filtration	*J Hazard Mater*	Materials / filtration	PM0.3 filtration efficiency, pressure drop	Computational / laboratory
Other clinical endpoints					
Sassano et al.	Household solid-fuel use and kidney disease mortality in the Golestan Cohort	*Environ Health Perspect*	Prospective cohort	Household fuel type and duration (biomass / kerosene / gas), questionnaire	Middle East & N. Africa

Provenance and method

Entry window **7 August 2026**, contiguous with the 6 August issue; no gap. Every PubMed record shipped here was verified by `efetch` to carry a `PubMedPubDate[@PubStatus="pubmed"]` of **7 August 2026** — 27 of 27, so no record was held for the next issue and none was re-indexed from an earlier window. Two Consensus records are dated January and February 2026 by Crossref `created` and are **explicitly flagged backfill** in their entries; they must be excluded from any weekly trend line. **Sources.** PubMed on two separate axes windowed on `[EDAT]` — the connector returned 19 health and 17 instrumentation hits (union 27), and **the local harvester returned 18 health and 0 sensing**, its sensing query producing an empty set for the second time. The connector remains the better recall leg on the instrumentation axis and both were merged before screening. Union 27, 1 dropped as an already-shipped duplicate, 18 carried, 8 rejected out of scope (Arctic marine particulate organic matter, indoor VOC fabric-air partitioning, cigarette-smoke bacteriology, a Legionella water assay, cannabis analytical chemistry, heat-not-burn fire-safety testing and two editorials with no primary data). Europe PMC on `CREATION_DATE` returned 1, already in the PubMed set. **Crossref scoped by journal ISSN across eight aerosol and atmospheric titles PubMed does not index returned 10 — the most productive sensing leg again** — of which 6 were carried and 4 rejected out of scope (GNSS refractivity, frost-point hygrometry, thunderstorm polarimetric radar, and a pollen/allergic-rhinitis study excluded as aeroallergen rather than mass-based PM). Consensus on three axes (calibration and drift; PurpleAir/Plantower network siting; satellite–surface fusion) returned 9 hits capped at 3 per sweep by the free tier; **7 resolved to DOIs already in `seen.json`** and 2 were carried as backfill. arXiv returned 60 records, **none published on or after 1 August**, so the recency screen carried nothing. OpenAlex returned `HTTP 429` on top of the standing paid-plan block — that leg has now failed on every run since 28 July. **Screening.** Deduplication keys PMID → lowercased DOI → title signature, checked against `seen.json` *before* summarising; Consensus titles were resolved to DOIs through Crossref before screening, which is what caught the seven duplicates. **Four records are carried on metadata alone** — DOIs verify in Crossref but no abstract exists in Crossref, Europe PMC or Semantic Scholar — and are marked in their entries; **no summary was written from a title**. First-author surnames and DOIs were read from `efetch`, taking `ELocationID[@EIdType='doi']` before `ArticleIdList` so that no reference-list DOI is picked up. **Labels.** Subtopic, design, particle-metric and geography assignments are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. The sixteen effect estimates are transcribed verbatim; one is a co-pollutant (NO_2) rather than PM and is marked as such in the forest caption and the register. Jiang et al.'s PM_{10} length-of-stay RR of 1.63 (0.18–15.94) is deliberately excluded from the forest plot: its interval spans an order of magnitude and would compress every other estimate to a point. All DOI links resolve to the publisher of record and passed the automated DOI gate before this issue compiled. Abstracts remain the copyright of their publishers.